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1. Business System Summary

We have designed a system named **Integrated Farm Management System (IFMS)** which is a relational database that brings together the important information used in a modern farm. It helps to store and organize data about workers, equipment, suppliers, fields, weather, crop production and harvesting etc. By linking details from Farmers, Equipment, Weather Stations, Fields, Crops and Harvest, IFMS gives a clear overall view of what is happening on the farm. This makes it easier to make decisions, plan maintenance, and keep track of everything on the farm [1].

1.1. Detailed Business Idea

The IFMS is built around the core assets and activities of a farm. Some of the detailed business ideas are mentioned below: [2]

- ❖ **Personnel and Structure:** The system focuses on Farmer who manage one or more Farms. Each farm can have multiple Fields.
- ❖ **Asset and Maintenance Management:** Equipment is used across the farm and it goes through scheduled maintenance to ensure it works properly when needed. Furthermore, suppliers provide the necessary materials and services for farm operations.
- ❖ **Precision Agriculture:** WeatherStations are deployed in different areas to monitor environmental conditions like temperature, humidity, and rainfall, which directly impact field operations.
- ❖ **Field Operations and Production:** Each Field has specific operations logged against it, including IrrigationLog entries and PestReport incidents.

- ❖ **Data-Driven Insights:** The relational model allows for analyzing the correlation between weather data, pest incidents, irrigation, and final harvest yields for specific fields.

1.2. Work Objectives

The primary objective of IFMS is to design a robust RDBMS that enables:

- ❖ **Comprehensive Asset Tracking:** To monitor equipment usage, maintenance history, and associated costs.
- ❖ **Enhanced Operational Logging:** Keeps the detailed records of field-specific activities like irrigation and pest control.
- ❖ **Improved Traceability:** Link harvest yields directly back to the field and the conditions it experienced.
- ❖ **Informed Decision-Making:** Answer key business questions to optimize resource allocation and improve yields.

1.3. Key Business Questions that IFMS will answer

Here are some the business questions that IFMS will be answering:

- What is the total maintenance cost for a specific piece of equipment?
- Which fields had pest reports in the last month, and what treatments were used?
- What was the total harvest (in kilograms) per farm?
- List all irrigation activities for a specific field, including water volume used.
- Which supplier provided the equipment that required the most frequent maintenance?

2. Logical Schema (ERD)

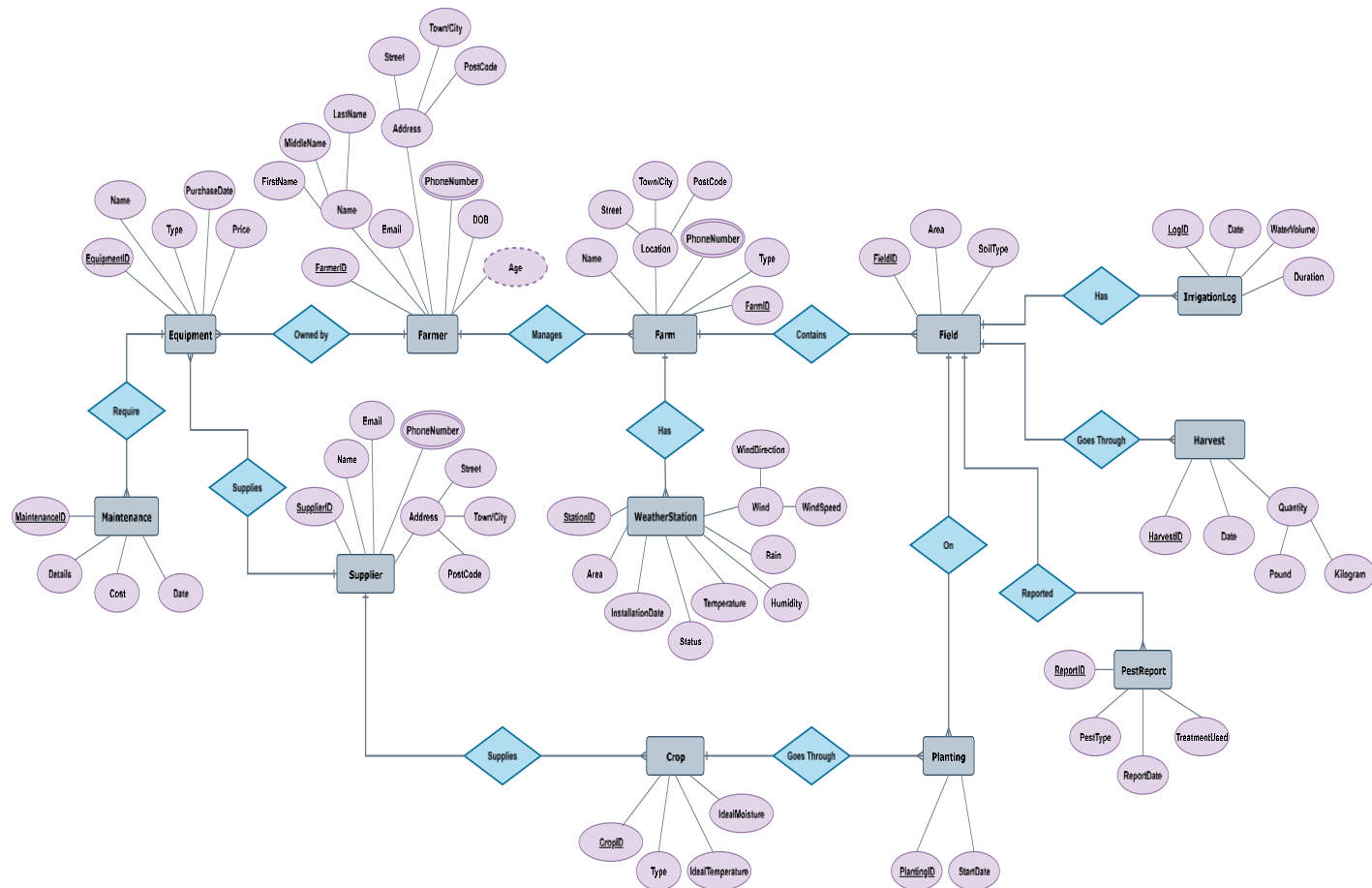


Figure 1: Entity-Relationship Diagram of IFMS

Above figure explains the logical schema of IFMS. Here we can see that IFMS consist of total 14 entities that are Equipment, Maintenance, Farmer, Farm, Field, Supplier, Crop, Planting, Harvest, Irrigation Log, Pest Report and Weather Station with multiple attributes. Each entity is partially dependent with another and shows relations between them. We can understand the system better with the help of table given below;

S.N.	Entity A	Entity B	Cardinality	Dependency	Relationship	Description
1.	Equipment	Farmer	N:1	Partial	Owned By	Many equipments are owned by single farm.
2.	Equipment	Maintenance	1:N	Partial	Requires	Single equipment requires multiple maintenance.
3.	Supplier	Equipment	1:N	Partial	Supplies	One supplier supplies many equipments.
4.	Farmer	Farm	1:N	Partial	Manages	One farmer manages many farms.
5.	Farm	WeatherStation	1:N	Partial	Has	One farm consists of many weather stations.
6.	Farm	Field	1:N	Partial	Contains	A farm is made of multiple fields.
7.	Field	IrrigationLog	1:N	Partial	Has	One field has multiple irrigation logs.
8.	Field	Harvest	1:N	Partial	Goes Through	Single field goes through multiple harvesting process.
9.	PestReport	Field	N:1	Partial	Reported	Many pests report can be reported in single field.

10.	Supplier	Crop	1:N	Partial	Supplies	One supplier supplies many crops.
11.	Crop	Planting	1:N	Partial	Goes Through	Single crop goes through multiple planting sessions.
12.	Planting	Field	N:1	Partial	On	Planting is done multiple times in single field.

Table 1: Tabular chart consisting of relations and cardinality between entities

3. Physical Data Model

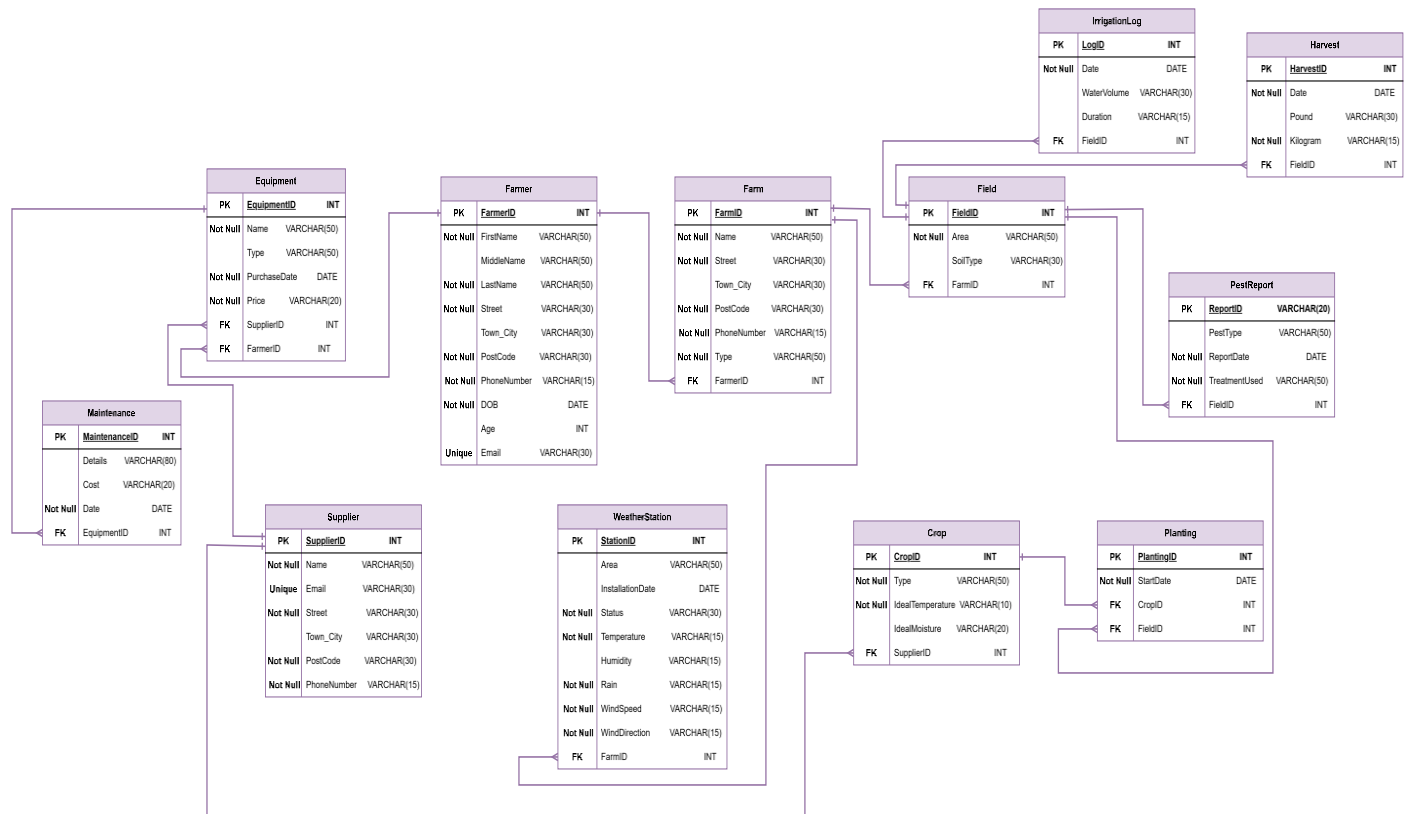


Figure 2: Physical Data Model of IFMS

Above diagram illustrates the physical data model of IFMS. Analyzing this diagram, we can see that there's a proper work flow between the system. Each table has been properly demonstrated by applying appropriate constraints and the keyed attributes has been highlighted along with other attributes. Proper data types have been used to ease the further process of development. Overall, this diagram cardinalities and foreign keys ensure correct relationships for example, one farm to many fields, one field to many harvests, and one supplier to many resources such as crop and equipment. And it shows that the system provides a well-organized database structure which supports efficient and accurate farm management.

4. References

- [1] Scribd, "SCRIBD," 2025. [Online]. Available: <https://www.scribd.com/document/850246285/Farm-Management-DBMS-Project-Report>.
- [2] BPlan, "Live Plan," 2025. [Online]. Available: <https://www.liveplan.com/sample-business-plans/technology/database-software-business-plan?ref=bplans>.